NORMALIZED VOLUMES ARE NOT LOWER SEMI-CONTINUOUS IN VALUATION SPACES

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Abstract. We show that the normalized volume function on the space of valuations centered at a point is not lower semi-continuous. This disproves a hypothesis of Li, which is also listed as an open question in Zhuang's survey.

1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let $x \in X$ be a klt singularity of dimension n and let $\mathrm{Val}_{X,x}$ be the space of real valuations of the function field $\mathbb{C}(X)$ that are trivial on $\mathbb{C}^*$ and centered at x, equipped with the weakest topology for which $v \to v(f)$ is continuous for any non-zero rational function f on X [JM12, Section 4.1]. For $v \in \mathrm{Val}_{X,x}$ with $A_X(v) < +\infty$, the *normalized volume* of v is

$$\widehat{\mathrm{vol}}_X(v) := A_X(v)^n \,\mathrm{vol}(v),$$

and $\widehat{\mathrm{vol}}_X(v) := +\infty$ if $A_X(v) = +\infty$, where $A_X(v)$ is the log discrepancy of v and $\mathrm{vol}(v)$ is the volume of v [Li18, Section 2.1].

Normalized volume is now a very useful tool in K-stability theory and birational geometry. Yet in the early days of the study of normalized volumes, the existence of a minimizer, i.e., a valuation v such that

$$\widehat{\mathrm{vol}}_X(v) = \inf_{w \in \mathrm{Val}_{X,x}} \widehat{\mathrm{vol}}_X(w),$$

was a central problem. Li [Li18, Hypothesis C on p. 501 and Conjecture 7.1(1)] expected that $\widehat{\mathrm{vol}}_X$ is lower semi-continuous on $\mathrm{Val}_{X,x}$, and showed that this implies the existence of a valuation minimizing $\widehat{\mathrm{vol}}_X$ [Li18, Corollary 4.3]. The existence of a minimizer was proved by Blum [Blu18], yet whether $\widehat{\mathrm{vol}}_X$ is lower semi-continuous on $\mathrm{Val}_{X,x}$ is still listed as an open question in [Zhu25, Question 6.20]. In this paper, we show that the answer is negative, already for smooth points.

Theorem 1.1. *Let $n \geq 3$ be an integer and let $o \in X := \mathbb{A}^n$ be the origin. Then $\widehat{\mathrm{vol}}_X$ is not lower semi-continuous on $\mathrm{Val}_{X,o}$. More precisely, there exist a valuation $w \in \mathrm{Val}_{X,o}$ and a sequence of valuations $v_1, v_2, \ldots$ in $\mathrm{Val}_{X,o}$ converging to w, such that*

$$\widehat{\mathrm{vol}}_X(v_j) = \frac{(2n-1)^n}{2^{n-1}} < \frac{(2n-2)^n}{2^{n-2}} = \widehat{\mathrm{vol}}_X(w)$$

for any j. In particular, [Li18, Hypothesis C] *fails and* [Zhu25, Question 6.20] *has a negative answer.*

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2. Proof of Theorem 1.1

We first recall some preliminaries. Notation and conditions as in Theorem 1.1. For $v \in \mathrm{Val}_{X,o}$ and $k > 0$, we let $\mathfrak{a}_k(v) := \{f \in \mathcal{O}_{X,o} \mid v(f) \geq k\}$ and

$$\mathrm{vol}(v) := \limsup_{k \to +\infty} \frac{\mathrm{length}(\mathcal{O}_{X,o}/\mathfrak{a}_k(v))}{k^n/n!}.$$

We refer the reader to [JM12, Section 5] for the log discrepancy $A_X(v)$.

Let $u_1, \ldots, u_n$ be linear coordinates on $\mathbb{A}^n$ and let $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{Z}^n_{>0}$. The *monomial valuation* with weights α in the coordinates $u_1, \ldots, u_n$ is the valuation $v \in \mathrm{Val}_{X,o}$ such that

$$v\left(\sum_{\beta \in \mathbb{Z}^n_{\geq 0}} c_\beta u_1^{\beta_1} \cdots u_n^{\beta_n}\right) = \min\left\{\sum_{i=1}^n \alpha_i \beta_i \;\middle|\; c_\beta \neq 0\right\}$$

for any non-zero polynomial $\sum_\beta c_\beta u_1^{\beta_1} \cdots u_n^{\beta_n}$. Then

$$A_X(v) = \sum_{i=1}^n \alpha_i \quad \text{and} \quad \mathrm{vol}(v) = \frac{1}{\prod_{i=1}^n \alpha_i}. \tag{2.1}$$

Indeed, the first equality follows from [JM12, Proposition 5.1] applied to the log smooth pair $(\mathbb{A}^n, \sum_{i=1}^n (u_i = 0))$. For the second equality, $\mathfrak{a}_k(v)$ is generated by the monomials $u_1^{\beta_1} \cdots u_n^{\beta_n}$ with $\sum_{i=1}^n \alpha_i \beta_i \geq k$, so $\mathrm{length}(\mathcal{O}_{X,o}/\mathfrak{a}_k(v))$ is the number of $\beta \in \mathbb{Z}^n_{\geq 0}$ with $\sum_{i=1}^n \alpha_i \beta_i < k$, which is

$$\frac{k^n}{n! \prod_{i=1}^n \alpha_i} + O(k^{n-1}).$$

Proof of Theorem 1.1. Let $x, y, z_3, \ldots, z_n$ be linear coordinates on $\mathbb{A}^n$. For any $a \in \mathbb{C}$, let v_a be the monomial valuation with weights $(1, 2, 2, \ldots, 2)$ in the coordinates $x, y - ax, z_3, \ldots, z_n$, and let w be the monomial valuation with weights $(1, 1, 2, \ldots, 2)$ in the coordinates $x, y, z_3, \ldots, z_n$. Fix pairwise distinct complex numbers $a_1, a_2, \ldots$ and let $v_j := v_{a_j}$.

Step 1. In this step we compute the normalized volumes. By (2.1),

$$A_X(v_a) = 2n - 1, \quad \mathrm{vol}(v_a) = \frac{1}{2^{n-1}}, \quad A_X(w) = 2n - 2, \quad \mathrm{vol}(w) = \frac{1}{2^{n-2}}$$

for any $a \in \mathbb{C}$, so $\widehat{\mathrm{vol}}_X(v_a) = \frac{(2n-1)^n}{2^{n-1}}$ and $\widehat{\mathrm{vol}}_X(w) = \frac{(2n-2)^n}{2^{n-2}}$. We have $\widehat{\mathrm{vol}}_X(v_a) < \widehat{\mathrm{vol}}_X(w)$ if and only if $\left(1 + \frac{1}{2n-2}\right)^n < 2$. If $n = 3$, then $\left(\frac{5}{4}\right)^3 = \frac{125}{64} < 2$. If $n \geq 4$, then

$$\left(1 + \frac{1}{2n-2}\right)^n < e^{\frac{n}{2n-2}} \leq e^{\frac{2}{3}} < 2,$$

where the last inequality holds as $e^2 < 8$.

Step 2. In this step we show that v_j converges to w. Since $v_j(g/h) = v_j(g) - v_j(h)$ and $w(g/h) = w(g) - w(h)$, it suffices to show that $v_j(f) = w(f)$ for any non-zero polynomial f and any $j \gg 0$.

Give x, y weight 1 and $z_3, \ldots, z_n$ weight 2, and write $f = \sum_{d \geq w(f)} f_d$, where f_d is homogeneous of weighted degree d and $f_{w(f)} \neq 0$. Fix $a \in \mathbb{C}$ and let $y_a := y - ax$. Since $y = y_a + ax$, each f_d is also homogeneous of weighted degree d as a polynomial in $x, y_a, z_3, \ldots, z_n$, where y_a has weight 1. Thus w is the monomial valuation with weights $(1, 1, 2, \ldots, 2)$ in the coordinates $x, y_a, z_3, \ldots, z_n$, and by comparing the weights of the monomials, we have $v_a(f_d) \geq d$ for any d, with equality if and only if some monomial of f_d in $x, y_a, z_3, \ldots, z_n$ is not divisible by y_a, that is, if and only if $f_d(x, ax, z_3, \ldots, z_n) \neq 0$. Hence $v_a(f) = w(f)$ if $f_{w(f)}(x, ax, z_3, \ldots, z_n) \neq 0$.

Consider the polynomial $F(t, x, z_3, \ldots, z_n) := f_{w(f)}(x, tx, z_3, \ldots, z_n)$. Since the morphism $\mathbb{A}^n \to \mathbb{A}^n$ given by $(t, x, z_3, \ldots, z_n) \to (x, tx, z_3, \ldots, z_n)$ is dominant and $f_{w(f)} \neq 0$, we have

$F \neq 0$. Thus $F(a, x, z_3, \dots, z_n) \neq 0$ for all but finitely many $a \in \mathbb{C}$. Since $a_1, a_2, \dots$ are pairwise distinct, $v_j(f) = w(f)$ for any $j \gg 0$.

Step 3. We conclude the proof. Let $c := \frac{(2n-1)^n}{2^{n-1}}$. By **Step 1**, $\widehat{\mathrm{vol}}_X(w) > c$. If $\widehat{\mathrm{vol}}_X$ were lower semi-continuous, then $\{v \in \mathrm{Val}_{X,o} \mid \widehat{\mathrm{vol}}_X(v) > c\}$ would be an open neighborhood of w, so it would contain v_j for any $j \gg 0$ by **Step 2**. This is not possible as $\widehat{\mathrm{vol}}_X(v_j) = c$ for any j by **Step 1**. □

Remark 2.1. The function A_X is lower semi-continuous on $\mathrm{Val}_{X,o}$ [JM12, Lemma 5.7], and indeed $A_X(w) = 2n - 2 < 2n - 1 = A_X(v_j)$. Thus the failure of lower semi-continuity of $\widehat{\mathrm{vol}}_X$ comes from vol, as $\mathrm{vol}(w) = 2\,\mathrm{vol}(v_j)$ for any j.

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